

Options :

6406532333208. ✓ I have written answers on the answer sheets

6406532333209. ✖ Not applicable

Industry 4.0

Section Id :	64065349300
Section Number :	11
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	25
Number of Questions to be attempted :	25
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653103543
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 215 Question Id : 640653698644 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532333210. ✓ YES

6406532333211. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653103544
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 216 Question Id : 640653698645 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following functions, Logistics Management does not cover?

Options :

6406532333212. ✗ Supply Management

6406532333213. ✗ Materials Management

6406532333214. ✓ Customer Management

6406532333215. ✗ Distribution Management

Question Number : 217 Question Id : 640653698647 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What is Last Mile in a supply chain?

Options :

6406532333219. ✖ Movements from Vendor to Distribution centres

6406532333220. ✖ Movements from Distribution centres to stores

6406532333221. ✔ Movements from stores to customers

Question Number : 218 Question Id : 640653698651 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What is the disadvantage of product layout?

Options :

6406532333231. ✖ Limited opportunities for the use of specialised equipment

6406532333232. ✖ High per unit cost for production

6406532333233. ✖ Movement of materials is tedious

6406532333234. ✔ Not very robust if there is a disruption

Question Number : 219 Question Id : 640653698654 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the correct progression of the industry?

Options :

6406532333240. ✖ Steam engine -> IT ->Assembly line ->IOT

6406532333241. ✔ Steam engine -> Assembly line -> IT -> IOT

6406532333242. ✖ Assembly line ->Steam engine -> IT ->IOT

Question Number : 220 Question Id : 640653698655 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

The Production rate of an assembly line is determined by its

Options :

6406532333243. ✔ Slowest station

6406532333244. ✖ Fastest station

6406532333245. ✖ Idle station

6406532333246. ✖ None of these

Question Number : 221 Question Id : 640653698660 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What is the decision variable X_{ij} denotes in a Vehicle Routing Problem (VRP)?

Options :

6406532333250. ✔ If the vehicle travels node j from node i (binary)

6406532333251. ✖ Number of vehicles travelled between node i and node j (integer)

6406532333252. ✖ Weigh on the arc (i,j) e.g. the distance (integer)

Question Number : 222 Question Id : 640653698664 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following examples can be referred as cyclic pattern?

Options :

6406532333260. ✔ Economic recession

6406532333261. ✖ Increases Ceiling Fan sales during winters

6406532333262. ✖ Increased tourist footfall towards the end of the year

Sub-Section Number : 3

Sub-Section Id : 640653103545

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 223 Question Id : 640653698646 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk powder packets in the neighbouring shelves. He picked a few milk powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sun-fist packets while he needed only two Good-day biscuits. Now map the scenarios with the items below:

Product	Scenario
P1)Tea:Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist bicuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406532333216. ✖ P1-S1, P2-S3, P3-S4

6406532333217. ✔ P1-S2, P2-S1, P3-S3

6406532333218. ✖ P1-S2, P2-S1, P3-S4

Question Number : 224 Question Id : 640653698662 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Select the correct form of internet from column B corresponding to the example in Column A.

Column A	Column B
A1. Smart metering	B1. Consumer
A2. Banking system	B2. Commercial
A3. Entertainment integrated devices	B3. Industrial
	B4. Private

Options :

6406532333254. ✖ A1: B3, A2: B2, A3:B4

6406532333255. ✔ A1: B3, A2: B2, A3:B1

6406532333256. ✖ A1: B2, A2: B3, A3:B1

Question Number : 225 Question Id : 640653698665 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following model corresponds to Exponential Smoothing forecasting model? [The notations represent their usual meanings]

Options :

$$F_{t+1} = \alpha D_t + (1 - \alpha)F_t \text{ where } F_t, D_t \text{ represents the forecast and actual}$$

6406532333263. ✓ data for period t

6406532333264. ✗ $F_t = \mu - \theta * \epsilon_{t-1}$ where $\epsilon_{t-1} = D_{t-1} - F_{t-1}$

Question Number : 226 Question Id : 640653698667 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is not an example of digitization?

Options :

6406532333270. ✗ Scanning a document to get the PDF document

6406532333271. ✗ Converting a physical 2D sheet to CAD drawing

6406532333272. ✗ Converting physical invoice to e-copy

6406532333273. ✓ Creating an automated report from the existing cost data

Question Number : 227 Question Id : 640653698668 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

You visit the car service centre to get your car washed. In the service centre, washing done using

automated systems, e.g. the car is put on a moving belt, water is sprayed, drying is also done.
Which is the following term representing this scenario?

Options :

6406532333274. ✖ Digitization

6406532333275. ✔ Automation

6406532333276. ✖ Digital transformation

Sub-Section Number :	4
Sub-Section Id :	640653103546
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 228 Question Id : 640653698648 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Choose the situation(s) from the options below in which a facility layout problem is relevant?

Options :

6406532333222. ✔ A new facility is added

6406532333223. ✔ A new product is introduced

6406532333224. ✔ A new service is introduced

6406532333225. ✔ Shift in technology

Question Number : 229 Question Id : 640653698666 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following supply chain costs is/are affected by distribution network structure?

Options :

6406532333265. ✔ Inventory

6406532333266. ✔ Transportation

6406532333267. ✔ Facilities and handling

6406532333268. ✔ Information

6406532333269. ✖ Price

Sub-Section Number :	5
Sub-Section Id :	640653103547
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 230 Question Id : 640653698649 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Short Answer Question

Calculate the Euclidean distance between two machines located at the coordinates A(5,10) and B(2,6) in a shop floor.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.3

Sub-Section Number :	6
Sub-Section Id :	640653103548
Question Shuffling Allowed :	Yes

Is Section Default? :	null
Question Number : 231 Question Id : 640653698650 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0	
Correct Marks : 1.5 Max. Selectable Options : 0	
Question Label : Multiple Select Question	
There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach.	
Options :	
6406532333227. ✓ (2,2)	
6406532333228. ✓ (3,2)	
6406532333229. ✓ (2.5,2)	
6406532333230. ✗ (2,3)	
Sub-Section Number :	7
Sub-Section Id :	640653103549
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 232 Question Id : 640653698652 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0	
Correct Marks : 1 Max. Selectable Options : 0	
Question Label : Multiple Select Question	
Process layout is also know as __ ? (select all that are applicable)	
Options :	
6406532333235. ✓ Functional layout	
6406532333236. ✓ Batch production layout	

6406532333237. ✖ Straight line layout

6406532333238. ✖ Curve layout

Question Number : 233 Question Id : 640653698663 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which are the two primary tasks of time series forecasting?

Options :

6406532333257. ✔ Identifying pattern

6406532333258. ✖ Identify variables of interest

6406532333259. ✔ Discovering relationships amongst variables of interest

Sub-Section Number :	8
Sub-Section Id :	640653103550
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 234 Question Id : 640653698653 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1.5

Question Label : Short Answer Question

A factory works for two 8 hour shifts each day and produces a total of 24,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. How many workstations are required to maintain the specified level of production (assuming that activities can be combined into those workstations in any order) (enter your answer in Integer value)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number : 9

Sub-Section Id : 640653103551

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653698656 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (235 to 237)

Question Label : Comprehension

Ramu puts up a stall for Diwali Light (Diya) every year. He builds a Diya at a cost of Rs. 3 and earns Rs. 5 by selling each Diya. Any Diya not sold by Diwali will be discounted to Rs. 1. From his experience, Ramu knows that the demand is Uniformly Distributed within [800, 1000] units. Then answer the given sub questions:

Sub questions

Question Number : 235 Question Id : 640653698657 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the Critical Ratio?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Number : 236 Question Id : 640653698658 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the optimal order quantity that minimizes Ramu's total cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

900

Question Number : 237 Question Id : 640653698659 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Percentage change in the order quantity if the demand support changes to [500,700]

Percentage change = $(Q_{\text{new}} - Q_{\text{old}}) / Q_{\text{old}}$

Enter your answer in percentage without '%' symbol and rounded off to zero decimal. E.g. if your answer is 0.1234 then enter your answer as 12

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

Sub-Section Number :

10

Sub-Section Id :

640653103552

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 238 Question Id : 640653698661 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Rana Bakery , a company is using a layout which consists of 9 workstations. 200 sandwiches are produced in a day on this layout. A typical day consists of a single shift which runs for 10 hours. The sandwich-making process consists of 9 activities (the information is provided in the table below).

Activity ID	Activity Description	Activity Time (in seconds)	Preceding activities
A	Wheat bag emptying	20	
B	Vegetable cleaning	10	
C	Bread baking	60	A
D	Vegetable cutting	40	B
E	Bread Slicing	20	C
F	Butter applying	5	E
G	Vegetable staffing	5	E, F
H	Sandwich baking	15	G
I	Sandwich packing	5	H

What is the current efficiency of the line where each activity is performed on a separate workstation (Enter only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” enter only “99.76”)?

Response Type :

Numeric

Evaluation Required For SA :

Yes

Show Word Count :

Yes

Answers Type :

Range

Text Areas :

PlainText

Possible Answers :

11.10 to 11.12

Sub-Section Number :	11
Sub-Section Id :	640653103553
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653698669 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Question Numbers : (239 to 241)

Question Label : Comprehension

Suppose, you are responsible for material purchase in an organization. You have demand forecast of a product for next four weeks which are 3, 4, 2 and 4 units respectively. You need to purchase one type of raw material to produce the final product to meet the above demand. Consider, one unit of the final product requires one unit of this raw material. Whenever an order is placed, your organization needs to incur a cost of \$4 irrespective of the quantity ordered. Purchasing cost of the raw material is \$2 per unit. Holding cost of an item not gone for production at the end of a week is \$0.5 per week. You can order a maximum of 5 units in a week and the warehouse has capacity to store upto 4 units of inventory in a week. Note that you need to meet the demand in every week.

Solve the above problem using dynamic programming (backward recursion) and answer the given sub questions for this problem.

Sub questions

Question Number : 239 Question Id : 640653698670 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Correct Marks : 2

Question Label : Multiple Choice Question

What does the state variable represent?

Options :

6406532333277. ✖ How many units of the raw material needs to be purchased in a week

6406532333278. ✖ Ending inventory with the warehouse in each week

6406532333279. ✔ Beginning inventory with the warehouse in each week

Question Number : 240 Question Id : 640653698671 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What does the action set represents?

Options :

6406532333280. ✔ How many units of the raw material needs to be purchased in a week

6406532333281. ✖ Ending inventory with the warehouse in each week

6406532333282. ✖ Beginning inventory with the warehouse in each week

Question Number : 241 Question Id : 640653698672 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

What is the objective function value in Stage-3 when the state variable is 2 and your action is 0?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Sub-Section Number :	12
Sub-Section Id :	640653103554
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653698673 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Question Numbers : (242 to 243)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There are total 100 seats in the aircraft. You have two classes of passengers - economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travellers follows a uniform distribution in [10,40]. Then answer the given subquestions:

Sub questions

Question Number : 242 Question Id : 640653698674 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Correct Marks : 1

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 243 Question Id : 640653698675 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0